

Chapter 2 — Discretisation

Finite element methods in 1D

MTH3340 · Numerical Methods for PDEs

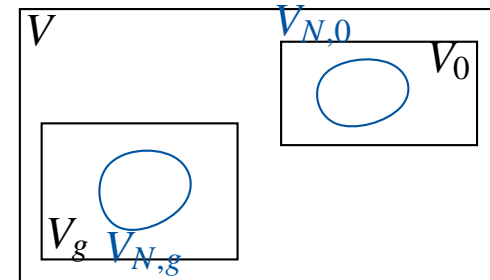
1 Where we are

The **Galerkin method** replaces V by a finite-dimensional space,

$$u_N \in V_{N,g} : a(u_N, v_N) = \ell(v_N) \quad \forall v_N \in V_{N,0},$$

with $\dim(V_N) = N < \infty$, **conforming**,

$$V_N \subset V, \quad V_{N,g} \subset V_g, \quad V_{N,0} \subset V_0,$$



which is what makes well-posedness (and the error analysis) free.

Spectral method: global polynomials, $V_N = \mathcal{P}_p(\Omega)$ on $\Omega = [a, b]$, $\dim(V_N) = p + 1$; on $[-1, 1]$, $V_{N,0} = \text{span}\{(1 - x^2)x^k\}_{k=0}^{p-2}$.

Q: Every basis function is supported on the whole domain, and Ω must be an interval. Can we do better?

2 Finite element method

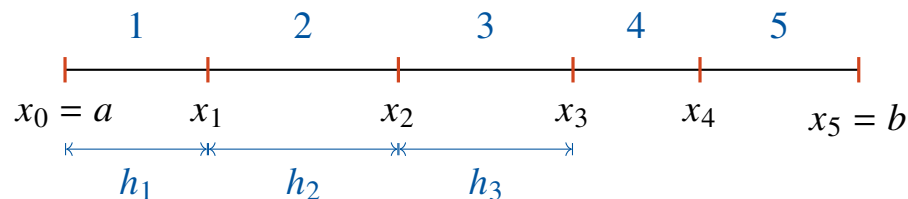
Idea: chop Ω into pieces and use polynomials *on each piece*, glued continuously. Piecewise polynomials instead of global ones. The first ingredient is the *mesh*.

Definition (Mesh in $\Omega \subset \mathbb{R}$)

Given $\Omega \doteq [a, b] \subset \mathbb{R}$, $M \in \mathbb{N}$ and a set of *nodes* $\Xi_M \doteq \{a \doteq x_0 < x_1 < \dots < x_{M-1} < x_M \doteq b\}$, the associated mesh is

$$\mathcal{M}_M \doteq \{(x_{j-1}, x_j) : 1 \leq j \leq M\}.$$

The open subintervals are the *cells*, of size $h_j \doteq |x_j - x_{j-1}|$; the mesh *width* is $h \doteq \max_{j=1, \dots, M} h_j$. If $h_j = h$ for all j , the mesh is *uniform*.



For a **uniform** mesh with M cells, $h = \frac{|b-a|}{M}$ and $x_i = a + i h$.

Hat functions

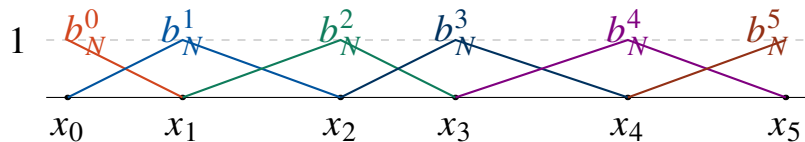
One basis function per node of $\mathcal{M}_{N+1}$ (N interior nodes).

Definition (Hat function in 1D linear finite elements)

For every interior node x_j , $j = 1, \dots, N$,

$$b_N^j(x) \doteq \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}}, & x_{j-1} \leq x < x_j, \\ 1 - \frac{x - x_j}{x_{j+1} - x_j}, & x_j \leq x < x_{j+1}, \\ 0 & \text{otherwise;} \end{cases}$$

at the end-points, $b_N^0(x) = 1 - \frac{x-a}{h_1}$ on $[a, x_1]$ and $b_N^{N+1}(x) = 1 - \frac{b-x}{h_{N+1}}$ on $[x_N, b]$ (zero elsewhere).



$b_N^j(x_i) = \delta_{ij}$, and b_N^j vanishes outside the two cells containing x_j : a **local** basis.

The finite element space

$$V_N \doteq \text{span}\{b_N^0, \underbrace{b_N^1, \dots, b_N^N}_{\text{interior}}, b_N^{N+1}\}, \quad \dim(V_N) = N+2,$$

$$V_{N,0} \doteq \text{span}\{b_N^1, \dots, b_N^N\}, \quad \dim(V_{N,0}) = N,$$

$$V_{N,g} = u_{N,0} + V_{N,0} \quad (\text{offset function, below}).$$

Functions in $V_{N,0}$ vanish at a and b .

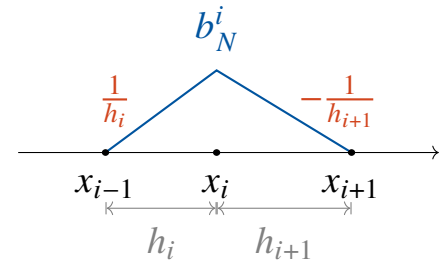
Equivalently, *without* mentioning a basis,

$$V_N = \{v \in C^0([a, b]) : v|_{[x_{j-1}, x_j]} \in \mathcal{P}_1, j = 1, \dots, N+1\},$$

$$V_{N,0} = V_N \cap C_0^0([a, b]).$$

The derivative is piecewise constant,

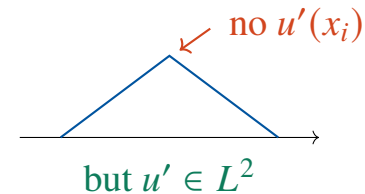
$$\frac{db_N^j}{dx}(x) = \frac{1}{h_j} \text{ on } (x_{j-1}, x_j), \quad -\frac{1}{h_{j+1}} \text{ on } (x_j, x_{j+1}), \quad 0 \text{ otherwise.}$$



Is it conforming?

Hat functions have a **kink** at the nodes, so they have **no classical derivative** there. Finite element spaces would be useless if we insisted on pointwise derivatives.

But we proved in Chapter 1 that continuous, piecewise smooth functions are in H^1 : the kink is harmless for the weak derivative. Hence



$$b_N^j \in H^1(\Omega) \implies V_N \subset H^1(\Omega) = V, \quad V_{N,0} \subset H_0^1(\Omega) = V_0,$$

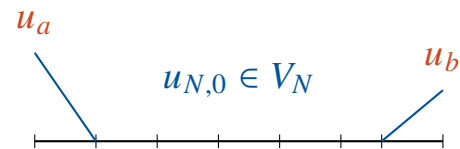
a **conforming** discretisation.

This is exactly why we spent Chapter 1 on weak derivatives: the variational formulation only asks for H^1 , and H^1 is what piecewise polynomials can deliver. As $N \rightarrow \infty$ (more, smaller cells) V_N grows and “ $V_N \rightarrow V$ ”. **How fast? Next lecture.**

Dirichlet conditions: the offset function

With $u(a) = u_a$, $u(b) = u_b$ and $a = x_0$, $b = x_{N+1}$: the two end-point hats were not used to build $V_{N,0}$; they are exactly what we need for the offset,

$$u_{N,0} \doteq u_a b_N^0 + u_b b_N^{N+1} \in V_N.$$



Indeed $u_{N,0}(a) = u_a$ and $u_{N,0}(b) = u_b$, since $b_N^i(x_j) = \delta_{ij}$.

Definition (Finite element method with offset function)

On a mesh $\mathcal{M}_{N+1}$, with the offset $u_{N,0}$ above, compute

$$\delta u_N \in V_{N,0} : a(\delta u_N, v_N) = \ell(v_N) - a(u_{N,0}, v_N) \quad \forall v_N \in V_{N,0},$$

and set $u_N \doteq u_{N,0} + \delta u_N$.

The offset is local: only two right-hand side entries change.

The linear system

Expand the unknown in the shape function basis of $V_{N,0}$, $\delta u_N = \sum_{i=1}^N \alpha_i b_N^i$, and test with each b_N^j (enough, by linearity):

$$\sum_{i=1}^N \underbrace{a(b_N^i, b_N^j)}_{A_{ji}} \alpha_i = \underbrace{\ell(b_N^j) - a(u_{N,0}, b_N^j)}_{\vec{f}_j}, \quad j = 1, \dots, N,$$

that is,

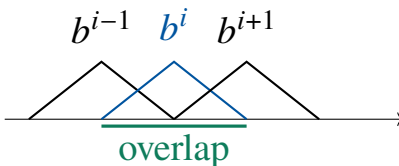
$$\mathbf{A} \vec{\alpha} = \vec{f}, \quad A_{ij} \doteq a(b_N^i, b_N^j), \quad \vec{f}_j \doteq \int_{\Omega} f b_N^j - a(u_{N,0}, b_N^j).$$

Same three steps as always: pick the space, pick the basis, assemble. What is new is the basis: **local support** $\Rightarrow a(b_N^i, b_N^j) = 0$ whenever the supports do not overlap.

The matrix of the 1D model problem

Example (1D model problem)

$u \in H_0^1((a, b))$ with $a(u, v) = \int_a^b u'v'$, $\ell(v) = \int_a^b f v$, homogeneous bc's and $\kappa = 1$. Compute A on a mesh of $N + 1$ cells.

$$A_{ij} = \int_a^b \frac{db_N^i}{dx} \frac{db_N^j}{dx} = \begin{cases} -\frac{1}{h_i}, & j = i - 1, \\ \frac{1}{h_i} + \frac{1}{h_{i+1}}, & j = i, \\ -\frac{1}{h_{i+1}}, & j = i + 1, \\ 0 & \text{otherwise,} \end{cases}$$


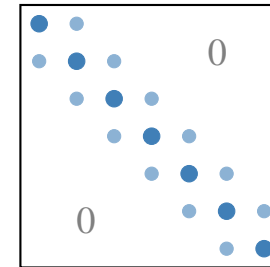
using that the derivatives are piecewise constant.

Sparsity: $A_{ij} = 0$ as soon as $|i - j| \geq 2$: neighbouring hats only. **The matrix is tridiagonal.**

Sparsity

$$\mathbf{A} = \begin{bmatrix} \frac{1}{h_1} + \frac{1}{h_2} & -\frac{1}{h_2} & & & \\ -\frac{1}{h_2} & \frac{1}{h_2} + \frac{1}{h_3} & -\frac{1}{h_3} & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{1}{h_N} & \frac{1}{h_N} + \frac{1}{h_{N+1}} & \\ & & & & \end{bmatrix}$$

$O(N)$ nonzeros instead of $O(N^2)$, and the same in 2D and 3D: each shape function only meets the few others sharing a cell with it.



Sparsity is the reason finite elements scale: storage and matrix products cost $O(N)$, and specialised solvers exploit the structure (a tridiagonal system is solved in $O(N)$ flops).

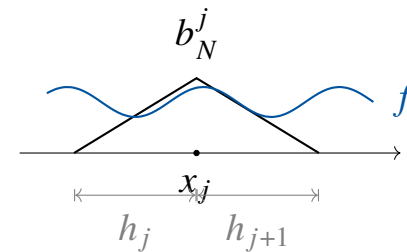
The right-hand side: numerical integration

$\vec{f}_j = \int_a^b f(x) b_N^j(x) dx$ cannot be computed exactly for a general f ; use a quadrature *cell by cell*. With the **trapezoidal rule**

$$\int_{x_{i-1}}^{x_i} g \approx \frac{h_i}{2} (g(x_{i-1}) + g(x_i)),$$

and since b_N^j vanishes at every node but x_j ,

$$\vec{f}_j \approx \frac{h_j + h_{j+1}}{2} f(x_j).$$



For $f \equiv 1$ this is exact: $\int_a^b b_N^j = \frac{h_j + h_{j+1}}{2}$.

The boundary conditions add $-a(u_{N,0}, b_N^j)$, which by locality is

$$-\int_a^b \frac{db_N^j}{dx} \frac{du_{N,0}}{dx} = \frac{u_a}{h_1} \quad (j = 1), \quad \frac{u_b}{h_{N+1}} \quad (j = N), \quad 0 \text{ (otherwise)}.$$

Uniform mesh: the classical system

For a uniform mesh of size h , homogeneous bc's and $f \equiv 1$,

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_N \end{bmatrix} = h^2 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}.$$

The finite element and finite difference systems coincide for a 1D uniform mesh with the trapezoidal rule on the right-hand side. **This is a coincidence of 1D uniform meshes; it fails for general meshes and for $d > 1$.**

The two methods are built differently, though. A difference stencil needs a **structured grid** to be written down at all, whereas the finite element space is assembled on an **arbitrary partition of Ω** : pick cells that fit the geometry, integrate over each of them. That is what will carry the construction to **unstructured meshes in 2D and 3D**.

Putting it together

Solving $A\vec{\alpha} = \vec{f}$ gives the coefficients of $\delta u_N \in V_{N,0}$, and

$$u_N = \underbrace{u_{N,0}}_{\text{bc's}} + \delta u_N$$

is a continuous piecewise linear function whose nodal values are $\alpha_0 = u_a$, $\alpha_1, \dots, \alpha_N$, $\alpha_{N+1} = u_b$.

As with the spectral method, we have built a **family** of spaces, now indexed by the mesh width,

$$V_{N,0}^h, \quad h \rightarrow 0 \ (N \rightarrow \infty), \quad u_N \longrightarrow u ?$$

Q: Does it converge, and at what rate in terms of h (equivalently, of the cost $N \simeq 1/h$)?

Error analysis and approximation theory, next.

